

JUnit Testing Exercises

Exercise 1: Setting Up JUnit

pom.xml dependency:

```
<dependency>
  <groupId>junit</groupId>
  <artifactId>junit</artifactId>
  <version>4.13.2</version>
  <scope>test</scope>
</dependency>
```

Sample Test Class:

```
import org.junit.Test;

public class SampleTest {
    @Test
    public void testSetup() {
        System.out.println("JUnit setup successful!");
    }
}
```

Expected Output:

JUnit setup successful!

Exercise 3: Assertions in JUnit

```
import static org.junit.Assert.*;
import org.junit.Test;
```

```
public class AssertionsTest {
    @Test
    public void testAssertions() {
        assertEquals(5, 2 + 3);
        assertTrue(5 > 3);
        assertFalse(5 < 3);
        assertNull(null);
        assertNotNull(new Object());
    }
}
```

Expected Output:

All test cases passed successfully.

Exercise 4: AAA Pattern, Setup and Teardown

```
import static org.junit.Assert.*;
import org.junit.*;
```

```
public class CalculatorTest {

    private int a, b;

    @Before
    public void setUp() {
        a = 10;
        b = 5;
        System.out.println("Setup completed");
    }

    @Test
    public void testAddition() {
        // Arrange: values initialized in setUp()
        // Act:
        int result = a + b;
        // Assert:
        assertEquals(15, result);
    }

    @After
    public void tearDown() {
        System.out.println("Teardown completed");
    }
}
```

}

Expected Output:
Setup completed
Test Passed
Teardown completed